RK200-02 Communication Protocol（MODBUS-RTU）

1.Communication parameters (factory default):

Baud rate: 9600bps, Data bits:8, Stop bit:1, Check bit:no, Address:0x09

2.Examples for read data

Host Scan Order：

Slave id	Function code	Address_H	Address_L	Quantity_H	Quantity_L	CRC_L	CRC_H
0x09	0x04	0x00	0x00	0x00	0x01	0X30	0x82

Sensor Response

Slave id	Function code	Number of bytes	Radiation (Unsigned integer 2 bytes)	CRC_L	CRC_H
0x09	0x04	0x02	0x0128	0X59	0X7F

Radiation value:(0128)H=(296)D=296(W/m^2)

3.Examples for modify the address (Take effect after restart)

Host Scan Order(change to 0x01)：

Slave id	Function code	Address_H	Address_L	New id_H	New id_L	CRC_L	CRC_H
0x09	0x06	0x00	0x42	0x00	0x01	0XE9	0x56

Sensor Response

Slave id	Function code	Address_H	Address_L	New id_H	New id_L	CRC_L	CRC_H
0x09	0x06	0x00	0x42	0x00	0x01	0XE9	0x56

Note: If you forget the original address, you should use the broadcast address(00H) (ensure that no other devices on the bus at this time.

4.Register description

Register name	Function code	Register address	Data type	Data length（byte）	R/W	Remark
Radiation	0x04	0x0000	Unsigned integer	2	R	**W/m^2**
Modify Sensor address	0x06	0x0042	Unsigned integer	1	W	0-0xFF
Baud rate	0x06	0x0040	Unsigned integer	1	W	0:1200 1:2400 2:4800 3:9600 4:19200 5:38400 6:57600 7:115200